Does Tokenization Fragmentation Predict Miscalibration in Bangla LLMs? A Cross-Lingual Study

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Abstract.

Keywords: Large language models · Bangla · Tokenization · Fragmentation · Confidence calibration · Cross-lingual evaluation

1 Introduction

1.1 Background and Motivation

1.2 Problem Statement

1.3 Research Gap

1.4 Research Questions and Contributions

- **RQ1:** How does tokenization fragmentation differ between Bangla and English under the selected model tokenizers?
- **RQ2:** Is item-level tokenization fragmentation associated with confidence and calibration behaviour on Bangla evaluation items?
- **RQ3:** Does this association vary across models, subjects, or language conditions?

1.5 Paper Organization

2 Related Work

2.1 Subword Tokenization and Multilingual Tokenizer Behaviour

2.2 Measuring Tokenization Fragmentation

2.3 Bangla LLMs and Bangla Benchmarking

2.4 Confidence Estimation and Calibration in LLMs

2.5 Summary of the Gap

3 Study Design and Research Questions

3.1 Study Overview

3.2 Hypotheses

4 Models and Datasets

4.1 Model and Tokenizer Selection

4.2 Dataset

For our study, we use the BnMMLU: Measuring Massive Multitask Language Understanding in Bengali [**?**] dataset. The dataset contains multiple-choice questions across the 23 domains, including science, humanities, mathematics, and general knowledge. It has primarily two categories: full, and hard.

The full category contains the samples which is simpler for large language models (LLMs) to answer, where as the hard category contains the samples which is difficult for the large language model to answer. Also, the hard category samples contains one extra feature named failure_ratio, which highlights the failure on large language model during the study.

Since, our study address the cross-lingual study, that is why we translated the dataset into English. Below, we provide the full character of the dataset and translation procedures.

Unique_Serial	subject_name	question	correct_answer	options	q
1	Algebra & Number Theory	(x+a)/(x+1)(x-3) = B/(x+1) + 1/(x-3) , A	B	-	

Table 1.

Dataset Characteristics

Translation Procedures of the Dataset

5 Tokenization Fragmentation Measurement

5.1 Text Unit Being Tokenized

5.2 Tokenizer Execution and Reproducibility

5.3 Word Counting and Normalization

5.4 Tokens per Word (TPW)

$$\mathrm{TPW}(x) = \frac{N_{\mathrm{tokens}}(x)}{N_{\mathrm{words}}(x)}. \tag{1}$$

5.5 Subword Token Retention Ratio (STRR)

5.6 Worked Tokenization Examples

6 Confidence and Calibration Methodology

6.1 Prompt and Multiple-Choice Scoring Protocol

6.2 Confidence Definition

6.3 Accuracy

$$\mathrm{Accuracy} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}(\hat{y}_i = y_i). \tag{2}$$

6.4 Brier Score

$$\mathrm{Brier} = \frac{1}{N}\sum_{i=1}^{N}\sum_{k=1}^{K}(p_{ik} - y_{ik})^2. \tag{3}$$

6.5 Expected Calibration Error (ECE)

$$\mathrm{ECE} = \sum_{b=1}^{B} \frac{|I_b|}{N} \left|\mathrm{acc}(I_b) - \mathrm{conf}(I_b)\right|. \tag{4}$$

6.6 Reliability Diagrams

6.7 Calibration Variants and Data Leakage Prevention

7 Experimental Setup

7.1 Evaluation Pipeline

7.2 Primary and Secondary Outcomes

7.3 Statistical Analysis Plan

7.4 Robustness and Sensitivity Checks

8 Results

8.1 Dataset and Model Coverage

8.2 Tokenization Fragmentation by Language and Model

8.3 Accuracy and Calibration Performance

8.4 Association Between Fragmentation and Calibration

8.5 Robustness Checks

9 Discussion

9.1 Summary of Main Findings

9.2 Interpretation and Comparison with Prior Work

9.3 Implications for Bangla LLM Evaluation

10 Limitations and Threats to Validity

10.1 Construct Validity

10.2 Internal Validity

10.3 External Validity

10.4 Statistical Conclusion Validity

11 Conclusion and Future Work

Acknowledgments.

Disclosure of Interests. The authors declare that they have no competing interests relevant to this work.